



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

GRADE/GRAAD 12

TECHNICAL MATHEMATICS P2/TEGNIJSE WISKUNDE V2

2023

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

CODE/ KODE	EXPLANATION/VERDUIDELIKING
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Volgehoue akkuraatheid
I	Identity/Identiteit
M	Method/Metode
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for omitting units/Geen penalisering vir eenhede weggelaat nie
R	Rounding/Afronding
RE	Reason/Rede
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule
ST/RE	Statement with reason/Bewering met rede
F	Correct formula/Korrekte formule

**These marking guidelines consist of 26 pages.
Hierdie nasienriglyne bestaan uit 26 bladsye.**

NOTE:

- If a candidate answers a question **TWICE**, only mark the **FIRST** attempt.
- Consistent Accuracy marking must be applied where indicated.
- Questions where Tolerance Range will be applied: Q4.3; Q7.3.3; Q10.4 and Q11.2.2

LET WEL:

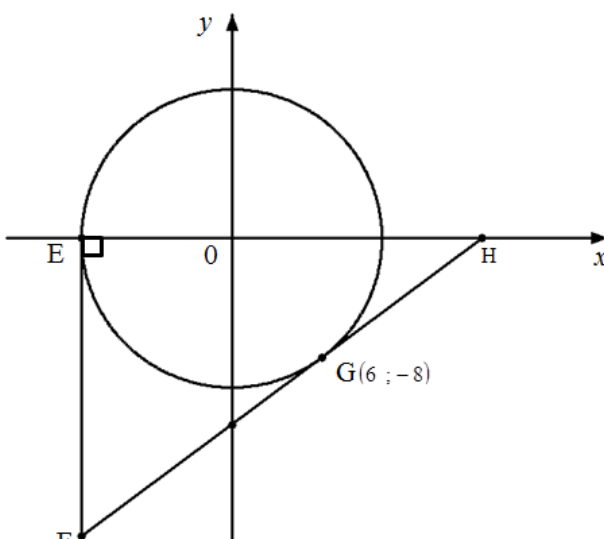
- Indien 'n kandidaat 'n vraag **TWEE** keer beantwoord, sien slegs die **EERSTE** poging na.
- Volgehoue akkuraatheid-nasien moet toegepas word soos aangedui.
- Vrae waar Tolaransie wydte toegepas word: V4.3; V7.3.3; V10.4 and V11.2.2

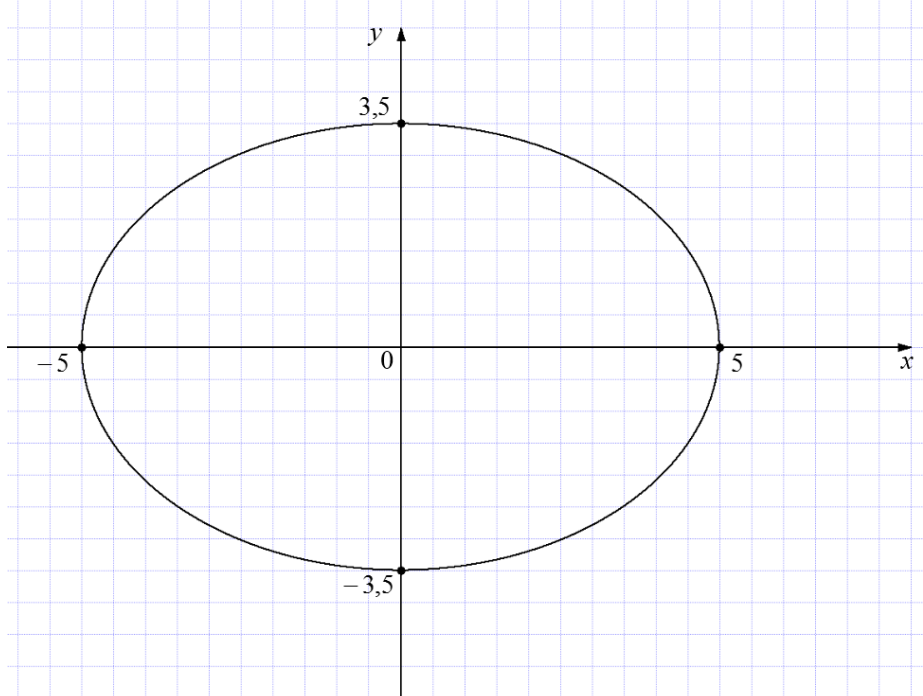
QUESTION/VRAAG 1

1.1	$m = 1$	✓ gradient/gradiënt	A (1)
1.2.1	$\tan \beta = m = 1$ $\therefore \beta = 45^\circ$	✓ SF ✓ value of angle/waarde van hoek AO: full marks/ volpunte	CA CA (2)
1.2.2	$y = x + 6$ OR/OF $y - x = 6$ $4 = d + 6$ OR/OF $4 - d = 6$ $\therefore d = -2$ OR/OF $\frac{4 - (-4)}{d - (-10)} = 1$ $8 = d + 10$ $d = -2$	✓ SF ✓ value of/ waarde van d OR/OF ✓ SF ✓ value of/ waarde van d AO: full marks/ volpunte	A CA CA CA (2)

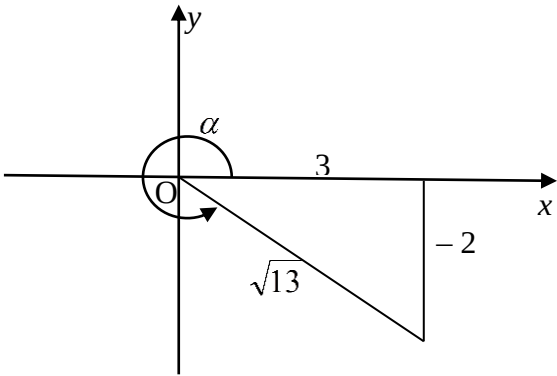
1.2.3	$P\left(\frac{x_k + x_m}{2}; \frac{y_k + y_m}{2}\right)$ $P\left(\frac{2 + (-10)}{2}; \frac{2 + (-4)}{2}\right)$ $P(-4; -1)$	✓ x-value/waarde A ✓ y-value /waarde A AO: full marks/ volpunte (2)
1.3.1	$m = 1$	✓ gradient/gradiënt CA From/ vanuit 1.1 (1)
1.3.2	$y = 1x + c$ $2 = 1(2) + c$ $\therefore c = 0$ $\therefore y = x$ OR/OF $y - y_1 = m(x - x_1)$ $y - 2 = 1(x - 2)$ $y - 2 = x - 2$ $\therefore y = x$	✓ substitution/ vervanging CA m from/van 1.3.1 and/en K(2; 2) ✓ equation/ vergelyking CA OR/OF ✓ substitution/ vervanging CA m from/van 1.3.1 and/en K(2; 2) ✓ equation/ vergelyking CA AO: full marks/ volpunte (2)
		[10]

QUESTION/VRAAG 2

			
2.1.1	$x^2 + y^2 = r^2$ $(6)^2 + (-8)^2 = r^2$ $x^2 + y^2 = 100$ OR/OF $y = \pm \sqrt{100 - x^2}$ OR/OF $x = \pm \sqrt{100 - y^2}$	✓ F A ✓ substitution/ vervanging A ✓ equation/ vergelyking CA AO: full marks/ volpunte (3)	
2.1.2	$m_{OG} = -\frac{8}{6} = -\frac{4}{3}$ $m_{\tan} = \frac{3}{4}$ $y = mx + c$ OR/OF $y - y_1 = m(x - x_1)$ $-8 = \frac{3}{4}(6) + c$ $y - (-8) = \frac{3}{4}(x - 6)$ $c = -\frac{25}{2}$ $\therefore y = \frac{3}{4}x - \frac{25}{2}$ OR/OF $y \cdot y_1 + x \cdot x_1 = r^2$ $y(-8) + x(6) = 100$ $-8y + 6x = 100$ $\therefore y = \frac{3}{4}x - \frac{25}{2}$	✓ SF A ✓ m_{OG} CA ✓ $m_{\tan}$ CA ✓ substitution/vervanging CA ✓ equation/vergelyking CA OR/OF ✓ F A ✓ substitution/verv (6;-8) A ✓ substitution/verv r^2 CA ✓ S CA ✓ equation/vergelyking CA (5)	

2.2	$E(-10; 0)$	✓ coordinates/ koördinate CA (1)
2.3	$y = \frac{3}{4}x - \frac{25}{2}$ $y = \frac{3}{4}(-10) - \frac{25}{2}$ $= -20$ $EF = 20 \text{ units}$ $EG = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(-10 - 6)^2 + (0 - (-8))^2}$ $= \sqrt{320} \text{ OR/OF } 8\sqrt{5} \approx 17,89 \text{ units / eenhede}$ $\therefore EF \neq EG$	✓ substitution /Vervang CA ✓ length/ lengte EF CA ✓ SF CA ✓ length/ lengte EG CA ✓ conclusion/gevolgtr CA (5)
2.4		✓ both x-intercepts/ beide x-afsnitte A ✓ both y-intercepts beide y-afsnitte A ✓ elliptical shape/eliptiese vorm CA (3) [17]

QUESTION/VRAAG 3

3.1.1	$\sin B$ $= \sin 59^\circ$ $\approx 0,86$ NPR	$\checkmark \approx 0,86$ A (1)
3.1.2	$\sec A + \cos B$ $= \sec 66^\circ + \cos 59^\circ$ $= \frac{1}{\cos 66^\circ} + \cos 59^\circ$ $\approx 2,97$ NPR	$\checkmark$ substitution/ vervanging A $\checkmark 2,97$ CA AO: full marks/volpunte (2)
3.2	$r^2 = x^2 + y^2$ $r^2 = (3)^2 + (-2)^2$ $r = \sqrt{13}$  $4 \cot \alpha + \sin^2 \alpha$ $= 4 \left(-\frac{3}{2} \right) + \left(-\frac{2}{\sqrt{13}} \right)^2$ $= -\frac{74}{13}$	$\checkmark r$ value/ waarde van A $\checkmark$ diagram A diagram can be implied by the use of $y = -2$ /diagram kan deur die gebruik van $y = -2$ geïmpliseer word. $\checkmark$ cot ratio/verhouding CA $\checkmark$ sin ratio/ verhouding CA $\checkmark$ simplification/ vereenv CA (5)
3.3	$\operatorname{cosec} x = -3,054$ $\frac{1}{\sin x} = -3,054$ $\sin x = -\frac{1}{3,054}$ OR / OF $\sin x \approx 0,327...$ Ref/verw.. $\angle \approx 19,11^\circ$ $x \approx 180^\circ + 19,11^\circ$ or/of $x \approx 360^\circ - 19,11^\circ$ $\therefore x \approx 199,11^\circ$ or/of $x \approx 340,89^\circ$ NPR	$\checkmark$ I A $\checkmark \sin x$ the subject/ onderwerp A $\checkmark$ reference angle/ verw.hk CA $\checkmark x$ - values in 3rd quadr/ waarde in die 3de kwadr. CA $\checkmark x$ - values in 4th quadr/ waarde in die 4de kwadr. CA Accept negative ref. angle/ aanvaar negatiewe verw.hk (5)
		[13]

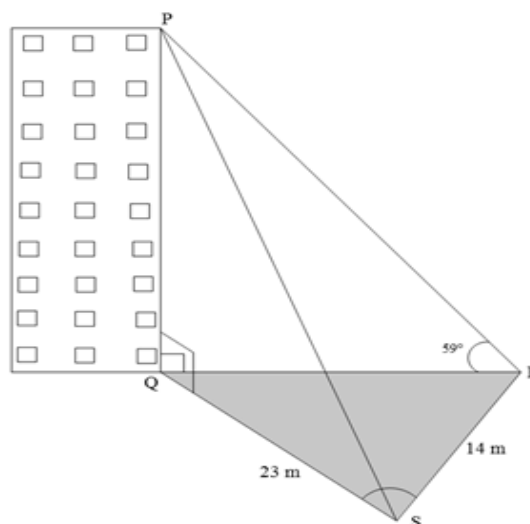
QUESTION/VRAAG 4

4.1	$\cos B$	$\checkmark \cos B$	A (1)
4.2.1	$\sin^2 P$	$\checkmark \sin^2 P$	A (1)
4.2.2	$\frac{\sin \beta}{\cos \beta}$ OR/OF $\frac{1}{\cot \beta}$ OR/OF $\frac{\pm \sqrt{\sec^2 \beta - 1}}{1}$	$\checkmark \mathbf{I}$	A (1)
4.3	$2 \sin(\pi + B) \cdot \cos(2\pi - B) \cdot \tan(180^\circ - B) + \cos(180^\circ - B) \cdot \frac{2}{\sec(180^\circ + B)}$ $= (-2 \sin B) \cdot \cos B \cdot (-\tan B) + (-\cos B) \cdot \frac{2}{(-\sec B)}$ $= 2 \sin B \cdot \cos B \cdot \frac{\sin B}{\cos B} + \cos B \cdot 2 \cos B$ $= 2 \sin^2 B + 2 \cos^2 B$ $= 2(\sin^2 B + \cos^2 B)$ $= 2(1)$ $= 2$	$\checkmark -2 \sin B$ $\checkmark -\tan B$ $\checkmark -\cos B$ $\checkmark -\sec B$ $\checkmark \mathbf{I} \cos B$ $\checkmark 2$	A A A A A CA (6)
4.4	$\frac{\cos \theta + \sin^2 \theta \cdot \sec \theta}{\operatorname{cosec} \theta} = \tan \theta$ $\text{LHS/ LK} = \frac{\cos \theta + \sin^2 \theta \cdot \sec \theta}{\operatorname{cosec} \theta}$ $= \frac{\cos \theta + \sin^2 \theta \cdot \frac{1}{\cos \theta}}{\frac{1}{\sin \theta}}$ $= \frac{\cos^2 \theta + \sin^2 \theta}{\cos \theta} \div \frac{1}{\sin \theta}$ $= \frac{1}{\cos \theta} \times \frac{\sin \theta}{1}$ $= \tan \theta = \text{RHS/RK}$	$\checkmark \mathbf{I} \frac{1}{\cos \theta}$ $\checkmark \mathbf{I} \frac{1}{\sin \theta}$ $\checkmark \mathbf{S} \cos^2 \theta + \sin^2 \theta$ $\checkmark \mathbf{I} 1$ $\checkmark \mathbf{S}$	A A CA A CA (5)
			[14]

QUESTION/VRAAG 5

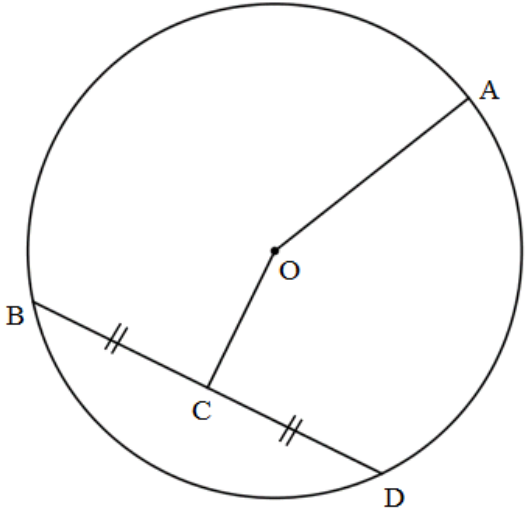
5.1		<i>f</i>: ✓ shape/vorm A ✓ intercept/afsnit A ✓ turning point/draaipunt A <i>g</i>: ✓ shape/vorm A ✓ intercepts/afsnitte A ✓ asymptote/asimptoot A (6)
5.2.1	$y \in [1; 2]$ OR/OF $1 \leq y \leq 2$ OR/OF $y \geq 1 \text{ and /en } y \leq 2$	✓ end points/eindpunte CA ✓ notation/notasie CA (2)
5.2.2	180°	✓ period/ periode A (1)
5.2.3	$h(x) = \tan x$	✓ equation/vergelyking A (1)
5.3	$= \sin 0^\circ + 1 - (-\tan 0^\circ)$ $= 1 - 0$ $= 1$	✓ Substitution / vervang A ✓ 1 CA AO: full marks/ volpunte (2)
		[12]

QUESTION/VRAAG 6



6.1	$QR^2 = QS^2 + SR^2 - 2QS \cdot SR \cos \hat{QSR}$ OR/OF $\text{in } \triangle QRS: s^2 = r^2 + q^2 - 2r \cdot q \cos \hat{S}$ $QR^2 = QS^2 + SR^2 - 2QS \cdot SR \cos \hat{S}$	$\checkmark 2QS \cdot SR \cos \hat{QSR}$ A (1)
6.2	$QR^2 = (23)^2 + (14)^2 - 2(23)(14)\cos 86^\circ$ $\approx 680,0768309$ $QR \approx 26,08 \text{ m}$	$\checkmark$ SF A $\checkmark$ value of/ waarde van CA QR NPR (2)
6.3	$\text{Area of/Oppervlakte van } \triangle QRS = \frac{1}{2} QS \times SR \sin \hat{S}$ OR/OF $\frac{1}{2} r q \sin \hat{S}$ OR/OF $\frac{1}{2} (23)(14) \sin \hat{S}$	$\checkmark QS \times SR$ OR/OF $r \times q$ A OR/OF 23×14 A (1)
6.4	$\text{Area of/oppervlakte van } \triangle QRS = \frac{1}{2} (23)(14) \sin 86^\circ$ $\approx 160,61 \text{ m}^2$	$\checkmark$ SF A $\checkmark$ area value/ oppervlakte waarde CA NPR (2)
6.5	$\tan 59^\circ = \frac{PQ}{QR}$ OR/OF $\frac{r}{p}$	$\checkmark \frac{PQ}{QR}$ OR/OF $\frac{r}{p}$ A (1)
6.6	$\tan 59^\circ \approx \frac{PQ}{26,08}$ OR/OF $\frac{PQ}{\sin 59^\circ} = \frac{26,08}{\sin 31^\circ}$ $PQ \approx 26,08 \tan 59^\circ \approx 43,40 \text{ m}$ $PQ \approx 43,40$	$\checkmark$ substitution/ vervanging CA $\checkmark$ value of/ waarde van CA PQ NPR (2)
		[9]

QUESTION/VRAAG 7

7.1	Bisects/ <i>halveer</i>	✓ST A (1)
7.2		
7.2.1	AO = 7 cm	✓ST A (1)
7.2.2	OC $\perp$ BD $\left(\begin{array}{l} \text{line from centre to midpt of chord} \\ \text{lyn vanuit midpt na midpt van koord} \end{array} \right)$ BC = 5,5 cm $OC^2 = BO^2 - BC^2$ (Pythagoras) $= 7^2 - 5,5^2$ $OC = \sqrt{18,75}$ $\approx 4,33$ cm OR/OF $4h^2 - 4dh + x^2 = 0$ $4h^2 - 4(14)h + (11)^2 = 0$ $4h^2 - 56h + 121 = 0$ $h = \frac{-(-56) \pm \sqrt{(56)^2 - 4(4)(121)}}{2(4)}$ $h \approx 11,33$ or / of $h \approx 2,67$ $\therefore OC \approx 7 - 2,67 \approx 4,33$ cm OR/OF	✓RE A ✓ST A ✓ST CA ✓ST length of/<i>lengte van</i> OC CA OR/OF ✓F A ✓SF A ✓Standard form/ <i>Standaard vorm</i> CA ✓ST length of/<i>lengte van</i> OC CA OR/OF

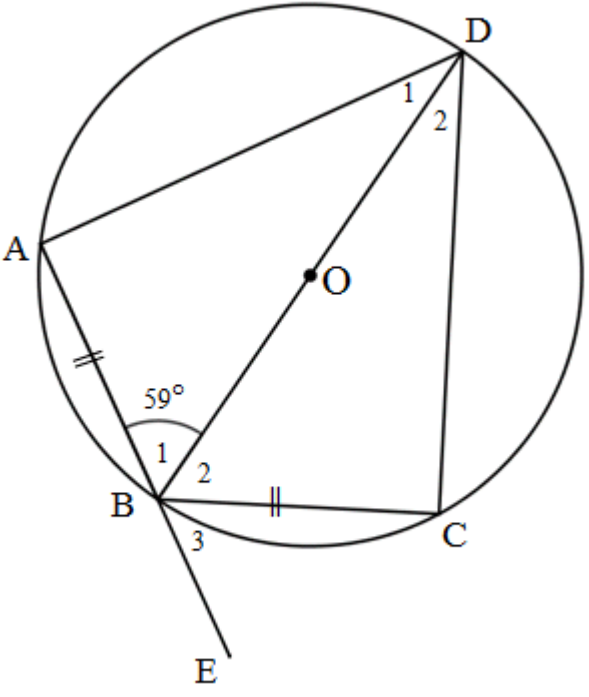
	$OC \perp BD$ (line from centre to midpt of chord) (lyn vanuit midpt na midpt koord) $BC = 5,5 \text{ cm}$ $\text{In } \triangle OBC, \cos \hat{OBC} = \frac{5,5}{7}$ $\hat{OBC} \approx 38,21^\circ$ $\tan 38,21^\circ = \frac{OC}{5,5}$ OR/OF $\sin 38,21^\circ = \frac{OC}{7}$ $OC \approx 4,33 \text{ cm}$ $OC \approx 4,33 \text{ cm}$	✓ RE A ✓ ST A ✓ cos definition/ def CA ✓ ST length of/lengte van OC (4)
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7.3			
7.3.1	$\hat{O}_1 = 4 \times 27^\circ = 108^\circ$	✓ ST	A (1)
7.3.2	$\hat{D}_1 = 54^\circ \left(\begin{array}{l} \angle \text{at centre} = 2 \times \angle \text{at circm} \\ \text{midpts } \angle = 2 \times \text{omtreks } \angle \end{array} \right)$	✓ ST ✓ RE	CA A (2)
7.3.3	$\hat{C}_2 = 54^\circ - 27^\circ = 27^\circ$ (ext $\angle$ of Δ / buite $\angle$ van Δ) $\therefore DC = DF$ (sides opp = $\angle$ s / sye teenoor = $\angle$ e) OR/OF $\hat{D}_2 = 180^\circ - 54^\circ = 126^\circ$ ($\angle$ on str line / op reguitlyn) $\hat{C}_2 = 180^\circ - 126^\circ - 27^\circ = 27^\circ$ (sum $\angle$ of Δ / som $\angle$ van Δ) $\therefore DC = DF$ (sides opp = $\angle$ s / sye teenoor = $\angle$ e)	✓ ST ✓ RE ✓ RE OR/OF ✓ ST CA ✓ RE A ✓ RE A (3)	
			[12]

QUESTION/VRAAG 8

8.1	Perpendicular / loodreg	✓ ST	A (1)
8.2			
8.2.1	$\hat{D}_1 = 55^\circ$ (tan - chord / raaklyn -koord))	✓ ST ✓ RE	A A (2)
8.2.2	$\hat{F}_3 = 55^\circ$ (alt/verw $\angle$ s; $CD \parallel FH$)	✓ ST ✓ RE	CA A (2)
8.2.3	$\hat{C}_3 = 35^\circ$ (tan/raaklyn $\perp$ rad) $\therefore \hat{F}_1 = 35^\circ$ ($\angle$ s opp=sides / $\angle$ e teenoor = sye) OR/OF $\hat{O}_1 = 110^\circ$ ($\angle$ at centre = $2 \times$ circmf $\angle$ / mdpts $\angle$ = $2 \times$ omtrks $\angle$) $\hat{F}_1 = \hat{C}_3 = \frac{1}{2} \times 70^\circ$ ($\angle$ s opp=sides / $\angle$ e teenoor = sye) $= 35^\circ$	✓ ST ✓ RE ✓ ST/RE ✓ ST ✓ RE ✓ ST/RE	A A CA A A CA (3)

8.2.4	$\hat{D}\hat{F}C = 180^\circ - 38^\circ - 35^\circ - 55^\circ \text{ (int } \angle\text{s of } \Delta \text{ / binne } \angle\text{e van } \Delta)$ $= 52^\circ \text{ (int } \angle\text{s of } \Delta \text{ / binne } \angle\text{e van } \Delta)$ OR/OF $\hat{C}_1 = 52^\circ \text{ (} \angle\text{on straight line / op reguitlyn)}$ $\hat{D}\hat{F}C = 52^\circ \text{ (tan-chord th. / raaklyn - koord)}$ OR/OF $\hat{C}_1 = 52^\circ \text{ (tan } \perp \text{ rad th. / raaklyn } \perp \text{ rad)}$ $\hat{D}\hat{F}C = 52^\circ \text{ (tan-chord th. / raaklyn - koord)}$	$\checkmark$ ST $\checkmark$ RE OR/OF $\checkmark$ ST $\checkmark$ RE OR/OF $\checkmark$ ST $\checkmark$ RE	CA A CA A CA A
			(2)

8.3			
8.3.1	$\hat{A} = 90^\circ$ ($\angle$ in semi-circle/sirkel) $\therefore \hat{D}_1 = 31^\circ$ (Int $\angle$ s of Δ / Binne $\angle$ evan Δ)	✓ ST ✓ RE ✓ ST	A A CA (3)
8.3.2	$\hat{D}_2 = 31^\circ$ (equal chords, equal $\angle$ s / <i>gelyke krde, gelyke $\angle$e</i>)	✓ ST ✓ RE	CA A (2)
8.3.3	$\hat{B}_3 = 62^\circ$ (ext $\angle$ of cyclic quad / buite $\angle$ van kdvk) OR/OF $\hat{B}_3 = 180^\circ - \hat{B}_2 - \hat{B}_1$ (sum $\angle$ on str line / som $\angle$ op reguitlyn) $= 180^\circ - 59^\circ - 59^\circ = 62^\circ$	✓ ST ✓ RE OR/OF ✓ ST ✓ RE	CA A CA A (2)
			[17]

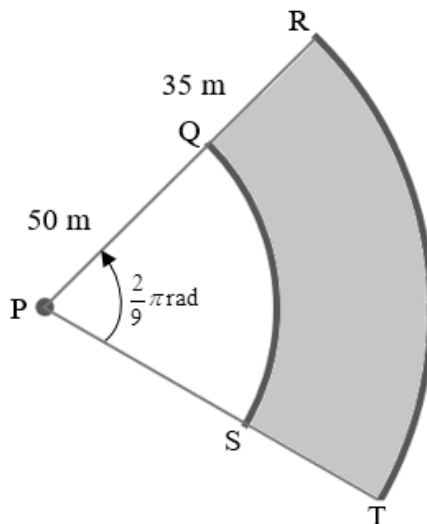
QUESTION/VRAAG 9

9.1	Corresponding angles must be equal/ ooreenstemmende hoeke moet gelyk wees Corresponding sides must be proportional/ Ooreenstemmende sye moet eweredig wees	✓ angles equal/ gelyke hoeke A ✓ sides proportional / sye eweredig A (2)
9.2	Piece/stuk 1 = $\frac{2}{5} \times 3\text{m} = 1,2\text{m}$ OR/OF = 120 cm Piece/stuk 2 = $3 - 1,2 = 1,8\text{m}$ OR/OF = 180 cm OR/OF Piece/stuk 2 = $\frac{3}{5} \times 300\text{ cm} = 180\text{ cm}$ OR/ OF $2x + 3x = 3\text{m}$ $x = 0,6\text{m}$ Piece/stuk 1 = $2x = 1,2\text{m}$ OR/OF = 120 cm Piece/stuk 2 = $3x = 1,8\text{m}$ OR/OF = 180 cm	✓ ST $\frac{2}{5} \times 3$ A ✓ ST length of piece 1/ lengte van stuk 1 CA ✓ ST length of piece 2 /lengte van stuk2 CA OR/OF ✓ ST $x = 0,6\text{ m}$ A ✓ ST length of piece 1/ lengte van stuk 1 CA ✓ ST length of piece 2 /lengte van stuk 2 CA (3)
9.3		
9.3.1	In $\triangle ABE$ and/en $\triangle DCE$: $\hat{A} = \hat{D}$ (alt./verw. $\angle$s; $AB \parallel CD$) $\hat{B} = \hat{C}$ (alt./verw. $\angle$s; $AB \parallel CD$) $\hat{E}_1 = \hat{E}_2$ (vertically opposite/regoorstaande $\angle$) $\therefore \triangle ABE \parallel \triangle DCE$ ($\angle \angle \angle$)	Any two/enige twee ✓ ST A ✓ ST A ✓ RE A (3)

9.3.2	$\frac{AE}{AD} = \frac{BE}{CB} \quad \text{prop. / eweredig } AB // CD$ $\frac{4}{9} = \frac{BE}{80}$ $\therefore BE = \frac{320}{9} \text{ mm } \mathbf{OR/OF} \approx 35,56 \text{ mm}$ OR/OF $\frac{BE}{CE} = \frac{AE}{DE} \quad (\parallel \Delta s)$ $\frac{BE}{80 - BE} = \frac{AE}{DE} = \frac{4}{5}$ $\therefore 5 BE = 320 - 4 BE$ $\therefore 9 BE = 320$ $\therefore BE = \frac{320}{9} \text{ mm } \mathbf{OR/OF} \approx 35,56 \text{ mm}$	✓ ST A ✓ RE A ✓ ST AD = 9 A ✓ ST CA OR/OF ✓ ST A ✓ RE A ✓ ST CE = 80 – BE A ✓ ST CA (4)
		[12]

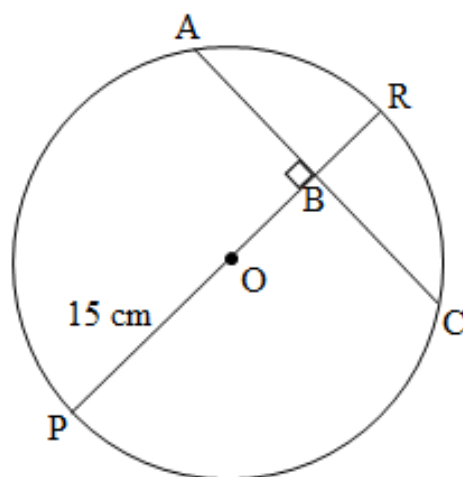
10.3.2	$\omega = 2\pi n$ $= 2\pi(3,06)$ $= \frac{153}{25}\pi \text{ OR / OF } \approx 19,23 \text{ rad / s}$ OR/OF $v = \omega r$ $12,5 = \omega(0,65)$ $\omega = \frac{12,5}{0,65} = \frac{250}{13} \text{ OR / OF } \approx 19,23 \text{ rad / s}$ OR/OF $\omega = 360^\circ \times n$ $= 360^\circ \times 3,06$ $= 1101,06 \times \frac{\pi}{180^\circ}$ $\omega \approx 19,23 \text{ rad / s}$	$\checkmark$ F A $\checkmark$ SF CA $\checkmark$ angular velocity/ <i>hoeksnelheid</i> CA OR/OF $\checkmark$ F A $\checkmark$ SF CA $\checkmark$ angular velocity/ <i>hoeksnelheid</i> CA OR/OF $\checkmark$ F A $\checkmark$ SF CA $\checkmark$ angular velocity/ <i>hoeksnelheid</i> CA (3)
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10.4



10.4	$A = \frac{r^2 \theta}{2}$ $A_1 = \frac{(50)^2 \left(\frac{2}{9} \pi \right)}{2} = \frac{2\,500}{9} \pi \text{ m}^2 \approx 872,66 \text{ m}^2$ $A_2 = \frac{(85)^2 \left(\frac{2}{9} \pi \right)}{2} = \frac{7\,225}{9} \pi \text{ m}^2 \approx 2522,00 \text{ m}^2$ $\text{Shaded area/Gearseerde oppervlakte} = \frac{7\,225}{9} \pi \text{ m}^2 - \frac{2\,500}{9} \pi \text{ m}^2$ $= 525\pi \approx 1\,649,34 \text{ m}^2$ $\therefore$ the area CANNOT be fully covered/ oppervlakte KAN NIE volledig bedek word NIE OR/OF $A = \frac{r^2 \theta}{2} \quad \text{OR/OF} \quad \frac{\theta}{360^\circ} \pi r^2 \text{ where } \theta = 40^\circ$ $= \frac{\left(\frac{2}{9} \pi \right) (85^2 - 50^2)}{2} \quad \text{OR/OF} \quad \frac{1}{9} \pi (85^2 - 50^2)$ $= 525\pi \approx 1\,649,34 \text{ m}^2$ $\therefore$ the area CANNOT be fully covered oppervlakte KAN NIE volledig bedek word NIE OR/OF	✓ F A ✓ SF 50 A ✓ SF 85 A ✓ Area/Oppervlakte CA ✓ Conclusion/ gevolgtrekking CA OR/OF ✓ F A ✓ SF 85 A ✓ SF 50 A ✓ Area/Oppervlakte CA ✓ Conclusion/ gevolgtrekking CA OR/OF
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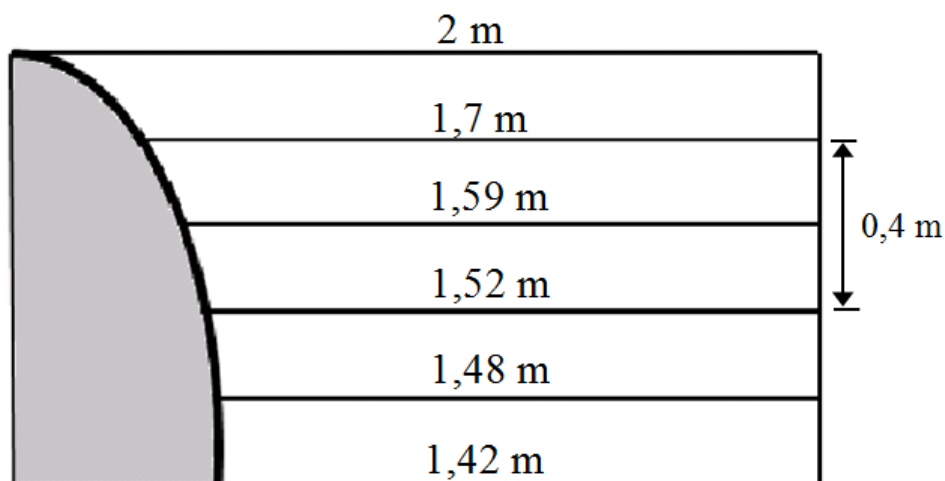
$s_1 = r\theta$ $= 50 \times \frac{2}{9} \pi = \frac{100}{9} \pi \text{ m} \approx 34,91 \text{ m}$ $A_1 = \frac{rs}{2}$ $= \frac{50 \times \frac{100}{9} \pi}{2} = \frac{2\,500}{9} \pi \text{ m}^2 \approx 872,66 \text{ m}^2$ $s_2 = r\theta$ $= 85 \times \frac{2}{9} \pi = \frac{170}{9} \pi \text{ m} \approx 59,34 \text{ m}$ $A_2 = \frac{rs}{2}$ $= \frac{85 \times \frac{170}{9} \pi}{2} = \frac{7\,225}{9} \pi \text{ m}^2 \approx 2522,00 \text{ m}^2$ $\therefore \text{Shaded/Gearseerde oppervlakte} = \frac{7\,225}{9} \pi \text{ m}^2 - \frac{2\,500}{9} \pi \text{ m}^2$ $= 525 \pi \text{ m}^2 \approx 1\,649,34 \text{ m}^2$ $\therefore$ the area CANNOT be fully covered <i>oppervlakte KAN NIE volledig bedek word NIE</i>	✓ F A ✓ SF 50 A ✓ SF 85 A ✓ Area/Oppervlakte CA ✓ Conclusion / <i>Gevolgtrekking</i> CA (5)
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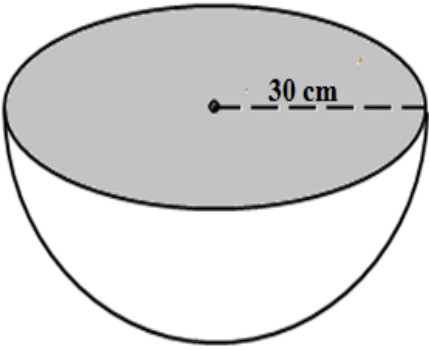
10.5	$4h^2 - 4dh + x^2 = 0$ $4h^2 - 4(30)h + (18)^2 = 0$ $4h^2 - 120h + 324 = 0$ $h = \frac{-(-120) \pm \sqrt{(120)^2 - 4(4)(324)}}{2(4)} \text{ OR / OF } (h - 27)(h - 3) = 0$ $h = 27 \text{ or / of } h = 3$ $\therefore BR = 3 \text{ cm}$ OR/OF Half chord method/ halfkoord metode: $AB = BC = 9 \text{ cm} \left(\begin{array}{l} \text{line from centre } \perp \text{ chord /} \\ \text{lyn vanuit mdpt } \perp \text{ koord} \end{array} \right)$ $AB^2 + OB^2 = OA^2 \text{ (Pythagoras Thm. / St.)}$ $9^2 + (15 - h)^2 = 15^2$ $81 + 15^2 - 30h + h^2 - 15^2 = 0$ $h^2 - 30h + 81 = 0$ $h = \frac{-(-30) \pm \sqrt{(30)^2 - 4(1)(81)}}{2(1)} \text{ OR / OF } (h - 27)(h - 3) = 0$ $h = 27 \text{ or / of } h = 3$ $\therefore BR = 3 \text{ cm}$	✓ F ✓ SF ✓ Standard form/<i>Standaard vorm</i> A A CA ✓ formula/factors/<i>formule faktore</i> CA ✓ length of/<i>lengte van</i> BR CA OR/OF ✓ length of/<i>lengte van</i> AB A ✓ SF ✓ Standard form/<i>Standaard vorm</i> A CA ✓ formula/factors/<i>formule faktore</i> CA ✓ length of/<i>lengte van</i> BR CA OR/OF
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	✓ length of/lengte van AB A OR/OF $AB=BC=9\text{ cm} \left(\begin{array}{l} \text{line from centre} \perp \text{chord/} \\ \text{lyn vanuit mdpt} \perp \text{koord} \end{array} \right)$ In $\triangle AOC$, $\sin \hat{AOC} = \frac{9}{15}$ $\therefore \hat{AOC} = 36,87^\circ$ $\tan \hat{AOB} = \frac{9}{OB}$ OR/OF $\cos \hat{AOB} = \frac{OB}{15}$ $OB = \frac{9}{\tan 36,87^\circ}$ $OB = 15 \cos 36,87^\circ$ $= 12\text{ cm}$ $= 12\text{ cm}$ $BR = 15 - OB$ $BR = 15 - OB$ $= 3\text{ cm}$ $= 3\text{ cm}$ OR/OF $AB=BC=9\text{ cm} \left(\begin{array}{l} \text{line from centre} \perp \text{chord/} \\ \text{lyn vanuit mdpt} \perp \text{koord} \end{array} \right)$ $AB^2 + OB^2 = OA^2$ (Pythagoras) $9^2 + OB^2 = 15^2$ $OB^2 = 15^2 - 9^2$ $OB^2 = 144$ $OB = 12\text{ cm}$ $BR = 15 - 12$ $\therefore BR = 3\text{ cm}$	✓ value of/waarde van $\hat{AOC}$ A ✓ ratio/verhouding CA ✓ length of/lengte van OB CA ✓ length of/lengte van BR CA OR/OF ✓ length of/lengte van AB A ✓ SF A ✓ length of/lengte van OB CA ✓ M CA ✓ length of/lengte van BR CA (5)
		[19]

QUESTION/VRAAG 11



11.1.1	0,2 m	✓0,2 m	A (1)
11.1.2	$A_T = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + \dots + o_{n-1} \right)$ $= 0,2 \left(\frac{2+1,42}{2} + 1,7 + 1,59 + 1,52 + 1,48 \right) \text{m}^2$ $= 0,2 (8) \text{m}^2$ $= 1,6 \text{ m}^2$	✓F	A
		✓SF	CA
		✓value of/ waarde van A _T	CA
	OR/OF $A_T = a (m_1 + m_2 + m_3 + \dots + m_n)$ $= 0,2 \left(\frac{2+1,7}{2} + \frac{1,7+1,59}{2} + \frac{1,59+1,52}{2} + \frac{1,52+1,48}{2} + \frac{1,48+1,42}{2} \right) \text{m}^2$ $= 0,2 (8) \text{m}^2$ $= 1,6 \text{ m}^2$		
			OR/OF
		✓F	A
		✓SF	CA
	✓value of/ waarde van A _T	CA (3)	

11.1.3	$4 = 0,2 (8 + 5x) \text{ m}^2$ $4 = 1,6 + x$ $4 - 1,6 = x$ $2,4 \text{ m} = x$ OR / OF $(x \times 0,2 \times 5) + 1,6 = 4 \text{ m}^2$ $x = 4 - 1,6$ $x = 2,4 \text{ m}$	✓ substitution of 5x/vervang 5x CA ✓ simplification/ vereenv CA ✓ value of / waarde van x CA OR / OF ✓ setting up equation/opstel van vergelyking CA ✓ simplification/ vereenv CA ✓ value of / waarde van x CA (3)
11.2		
		
11.2.1	Surface area of hemisphere/ <i>Buite opp. van hemisfeer</i> $= \frac{1}{2} \times 4\pi r^2$ $= 2\pi (30)^2$ $= 1\,800\pi$ OR / OF $\approx 5\,654,87 \text{ cm}^2$ OR/OF Surface area of sphere/ <i>Buite opp. van sfeer</i> $= 4\pi r^2$ $= 4\pi (30)^2$ $= 3\,600\pi \text{ cm}^2$ OR / OF $\approx 11\,309,73 \text{ cm}^2$ Surface area of hemisphere/ <i>Buite opp. van hemisfeer</i> $= \frac{3\,600\pi}{2} = 1\,800\pi$ OR/OF $\approx 5\,654,89 \text{ cm}^2$	✓ F A ✓ SF A ✓ surface.area/buite oppervlakte CA NPR OR / OF ✓ F A ✓ SF A ✓ surface area/buite oppervlakte CA NPR (3)

11.2.2	Volume of 75% hemisphere/ $= \frac{1}{2} \times \frac{4}{3} \pi r^3 \times 75\%$ <i>Volume van hemisfeer</i> $= \frac{1}{2} \times \frac{4}{3} \pi (30)^3 \times 0,75$ $= 13\,500\pi \text{ cm}^3$ OR / OF $\approx 42\,411,50 \text{ cm}^3$ Time taken to fill/ $= \frac{13500\pi}{90}$ <i>tyd geneem om te vul</i> $= 150\pi \text{ sec}$ OR/OF $\approx 471,24 \text{ s}$ $= 2,5\pi \text{ min}$ OR/OF $\approx 7,85 \text{ minutes}$ OR/OF Volume of sphere/van sfeer $= \frac{4}{3} \pi r^3$ $= \frac{4}{3} \pi (30)^3$ $= 36\,000\pi \text{ cm}^3$ OR / OF $\approx 113\,079,34 \text{ cm}^3$ Volume of 75% hemisphere/ $= \frac{36\,000\pi}{2} \times 75\%$ <i>Volume van hemisfeer</i> $= 13\,500\pi \text{ cm}^3$ OR/OF $\approx 42\,411,50 \text{ cm}^3$ time taken to fill/ $= \frac{13500\pi}{90}$ <i>tyd geneem om te vul</i> $= 150\pi \text{ s}$ OR/OF $\approx 471,24 \text{ s}$ $= 2,5\pi \text{ min}$ OR/OF $\approx 7,85 \text{ minutes}$	✓F A ✓SF A ✓ value of/ waarde van V CA ✓M (dividing /deel deer by 90) A ✓ time taken/tyd geneem CA NPR OR/OF ✓F A ✓SF A ✓ value of/ waarde van V CA ✓M (dividing deel deer by 90) A ✓ time taken/tyd geneem CA NPR (5)
		[15]

TOTAL/TOTAAL: 150